CS 2490 – Homework 5

Turn in by 2026/10/10 at 11:59pm

You **must** turn in your assignment as a single pdf file entitled: *lastname_firstname_hw5.pdf*. You can use any typesetting system (LaTeX, Word, etc.) or you can write your solutions by hand and scan them (make sure that your writing is legible).

Each of the exercises is about proving that a given language is not regular. Depending on the situation you can use the pumping technique seen in class, or you can use the properties of regular languages with respect with intersection, complement, etc. Or you might sometimes want to apply a combination of both! You can of course use all the results and examples seen in class without reproving them.

Exercise 1. In this exercise, the alphabet Σ can be any alphabet, but it has at least two letters. Show that

$$L = \{ww \mid w \in \Sigma^*\}$$

is not regular.

Exercise 2. Does the result of the previous exercise still hold if the alphabet Σ has only one letter? Justify your answer.

Exercise 3. In this exercise, the alphabet is $\Sigma = \{a\}$. Show that

$$L = \{a^n \mid n \text{ is prime}\}$$

is not regular.

Exercise 4. In this exercise, the alphabet is $\Sigma = \{a\}$. Show that

$$L = \{a^n \mid n \text{ is a power of } 2\}$$

is not regular.

Exercise 5. In this exercise, the alphabet is $\Sigma = \{a, b, c\}$. Show that

$$L = \{wcv \mid w \in \{a, b\}^* \wedge v \in \{a, b\}^* \wedge w \neq v\}$$

is not regular.